FRAMES OF REFERENCE, RELATIVITY AND TIME DILATION

LEARNING GOALS

Students will:

- Know the observations that led Einstein to develop the theory of special relativity.
- Know what is meant by time dilation and how to calculate it.

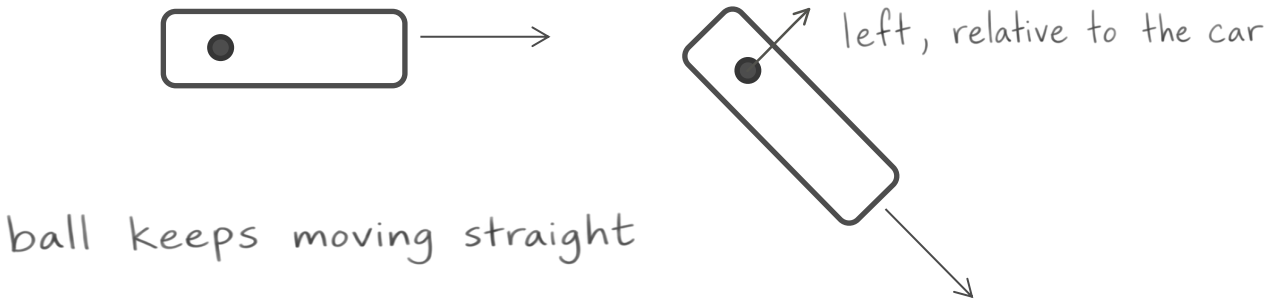
FRAMES OF REFERENCE

Recall: An inertial frame of reference is any frame of reference in which an object with zero net force remains at rest or moves in a straight line with constant speed.

EXAMPLE OF AN INERTIAL AND NON-INERTIAL FRAMES OF REFERENCE

A ball is sitting on a level table in a car that's moving directly north with a constant speed of 30 m/s. The ball remains at rest with respect to the car and remains moving at 30 m/s [N] with respect to the road.

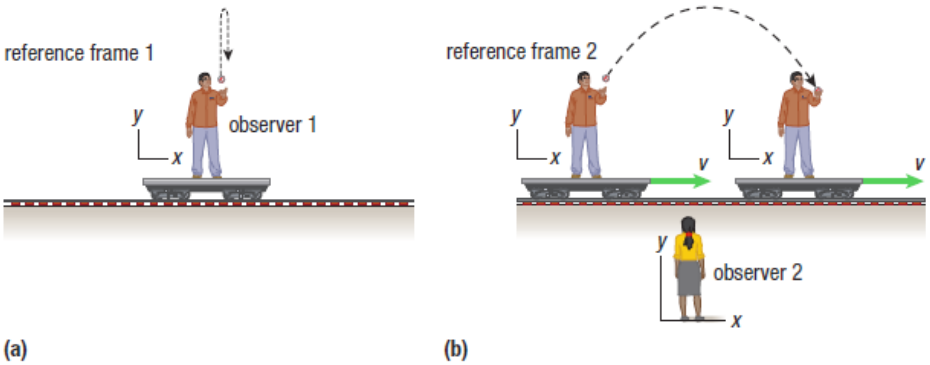
In a non-inertial frame of reference what happens to the ball if the car suddenly turns to the right? Draw the direction the ball moves with respect to the car.



The car has accelerated by changing directions and has now become a non-inertial frame of reference so with respect to the car, the ball now moves to the side and hits the car door. From the point of view of a passenger inside the car who is unaware of the acceleration, the ball appears to accelerate due to a net force applied to it by a fictitious force when in fact, it is only the car that accelerates.

EXAMPLE

(a) When observer 1 throws a ball upward, he observes that the ball's motion is purely along the vertical (y) direction.



(b) When the railway car for observer 1 has a speed *v* relative to observer 2, the ball appears to undergo projectile motion with a displacement along both *x* and *y* in reference frame 2. However, as viewed by observer 1 using his reference frame and coordinates *x* and *y*, the ball's motion is still purely vertical, as in (a).

PRINCIPLES OF SPECIAL RELATIVITY

The Principle of Relativity

- The laws of physics are the same in all inertial frames of reference. No physics experiment can ever determine whether you are at rest or moving at a constant velocity.

The Speed of Light Principle

- There is at least one inertial frame of reference in which, for an observer at rest in this frame of reference, the speed light, *c*, in a vacuum is independent of the motion of the source of the light.

THE SPEED OF LIGHT

The speed of light was first successfully measured by Michelson and Morley in 1887. Since then, the speed has been measured many times in many different circumstances.

If the rules we have learned for translating from one frame of reference to another work on light, we should be able to measure it. Imagine a friend holding a flashlight. She points it at you and turns it on. The beam travels from her to you at the speed of light $c = 3.00 \times 10^8$ m/s. Now imagine she is on a train that is travelling towards you at a speed v. She once again points the flashlight at you and turns it on. You would now expect the speed of the beam to travel towards you at a speed c+v. However, when scientists do such experiments, they find that the beam still appears to be travelling at a speed c. These experiments have been repeated many times and with great precision and scientists have found that the speed is always c. **For all observers in inertial frames of reference and all light sources in inertial frames of reference, the light travels from the source to the observer at the speed c regardless of the velocities of the frames of reference.** This has several implications which we will explore.

SIMULTANEITY

Two events that are simultaneous in one reference frame are not necessarily simultaneous in another reference frame that's moving relative to the first.

An observer O_s, stationary in the inertial frame of Earth, is standing on a railway platform at the midway point between two lampposts, L_1 and L_2 (Figure 5). The lampposts are connected to the same circuit, ensuring that, at least from the viewpoint of an inertial observer, they turn on at the same time.

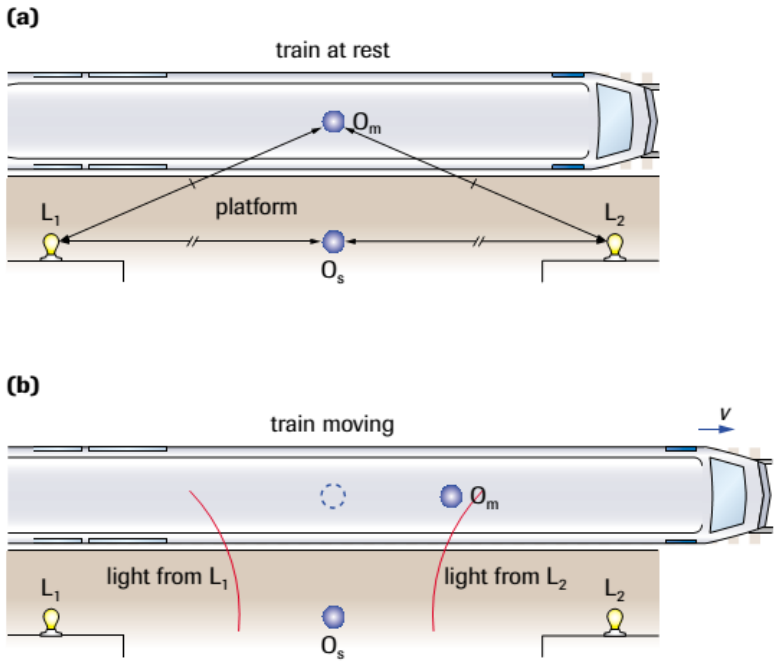


Figure 5

(a) When the train is at rest, each observer sees the lamps flash simultaneously, since each observer is halfway between the lamps.

(b) When the train is moving, each observer does not see the lamps flash simultaneously, since the light from L_1 takes longer to reach O_m than the light from L_2, that is $\Delta t_{L1} > \Delta t_{L2}$.

When the lights are turned on, the beams travel from the lights L_1 and L_2 equal distances to O_s and arrive at the observer at the same time.

Now, consider a moving observer O_m on a train travelling by the platform at some speed v. When O_m is exactly opposite O_s, the lights are turned on and beams leave L_1 and L_2. Since the train travels towards L_2, the distance from O_m to L_2 is decreasing as the beam travels. This beam will arrive at O_m first. At the same time, the distance from L_1 to O_m is increasing as the beam from L_1 travels. This beam will arrive at O_m after the beam from L_2. What the observer at O_s sees as a simultaneous event, the observer at O_m sees as happening at different times.

The interesting thing to note is that both observers' views are correct. How you perceive a sequence of events depends on your frame of reference. Of course, in everyday life these effects are so small that we never notice them. They only become noticeable when frames of reference are moving close to the speed of light.

Another explanation can be seen on youtube at the following link:
http://www.youtube.com/watch?v=wteiuxyqtoM

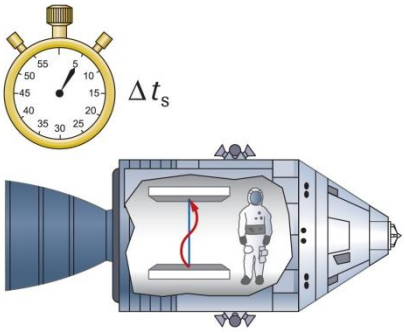
SPECIAL THEORY OF RELATIVITY

All physical laws are the same in all inertial frames of reference, and the speed of light is independent of the motion of the light source or its observer in all inertial frames of reference.

TIME DILATION

Another implication for the speed of light remaining constant for all inertial observers is that it means that observers in different reference frames **will disagree on distances and times**. In classical or Newtonian physics, we believe all observers in different inertial frames agree on the time elapsed for an event to occur. However, this can no longer be true if observers are to agree on the speed of light. And they MUST agree on the speed of light because THIS IS WHAT WE OBSERVE.

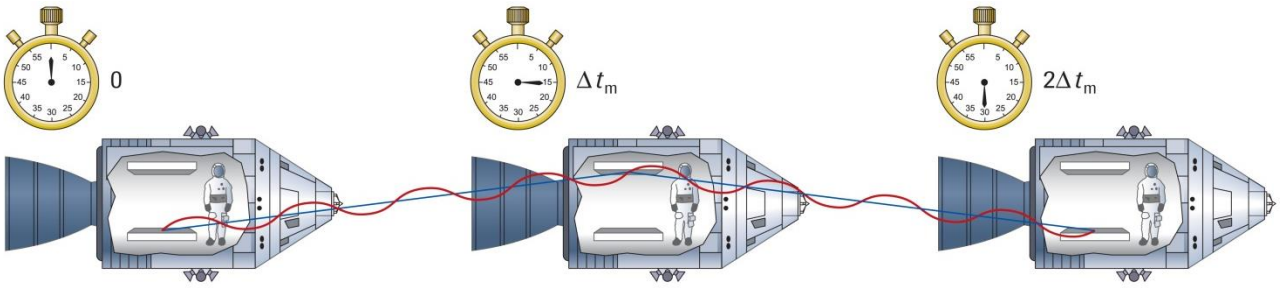
Therefore, all time is relative to the observer. There is no such thing as absolute time.



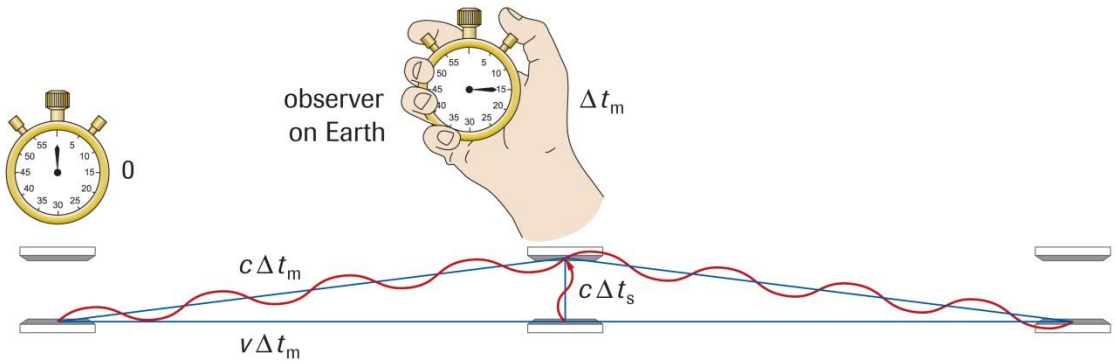
To understand this, consider two reference frames, a spaceship and Earth, in which a pulse of light is seen to travel from one mirror to another inside the spaceship.

According to the astronaut, the light pulse goes up and down whether the spaceship is at rest or moving with high constant velocity. The time it takes the light to travel from the top mirror to the bottom mirror is Δt_s, while travelling at the speed of light c. The distance the light travels is therefore,

$$\Delta d = c\Delta t_s$$



In the reference frame of the Earth, the spaceship is moving with a speed v and from an observer's point of view from the Earth, the pulse travels a longer interval of Δt_m over a longer distance $c\Delta t_m$. While the light travels this distance, the spaceship also travels a distance $v\Delta t_m$.



We can determine the relationship between Δt_s and Δt_m using the Pythagorean theorem.

$$(c\Delta t_m)^2 = (v\Delta t_m)^2 + (c\Delta t_s)^2$$

Isolate for Δt_m to see how it compares to Δt_s,

$$(c\Delta t_m)^2 - (v\Delta t_m)^2 = (c\Delta t_s)^2$$
$$c^2\Delta t_m^2 - v^2\Delta t_m^2 = c^2\Delta t_s^2$$

Divide both sides by c^2,

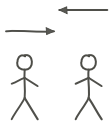
$$\Delta t_m^2 - \frac{v^2}{c^2}\Delta t_m^2 = \Delta t_s^2$$
$$\Delta t_m^2\left(1 - \frac{v^2}{c^2}\right) = \Delta t_s^2$$
$$\Delta t_m = \frac{\Delta t_s}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where $\boldsymbol{\Delta t_s}$ is the time interval for the observer **stationary relative to the sequence of events** and $\boldsymbol{\Delta t_m}$ is the time interval for an observer **moving with a speed v relative to the sequence of events**. Be careful how you interpret this. Many people would think the observer on Earth is stationary and the observer in the spaceship is moving. However, the events take place on the ship and the observer on the ship is the observer who is stationary with respect to the events.

THE TWIN PARADOX

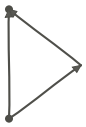
Read Page 573 on the twin paradox.

Explain why people who are beginners at relativity think there is a paradox.



Each twin sees the other moving, so each thinks the other twin should age less. Both cannot be younger when they meet again.

Explain why the two twins are not equivalent observers and therefore there is no paradox. Who is younger?



The travelling twin turns around and changes inertial frames. The Earth twin does not. The travelling twin is younger when they meet.

In our next lesson, you will learn how to calculate how much time elapses for each twin.

CLASS WORK

Page 568 Q1-4, Page 573 Q1-4